Recent advancements in scaling neural language models have prioritized empirical performance over foundational understanding, resulting in unpredictably fallible systems. Current methods for improving reliability perform well on visible benchmarks but often fail to generalize to slight variations of the original problem. For example, training models on human preferences to refuse certain inputs mitigates overtly malicious queries, but falls short when the same intent is subtly rephrased or framed.

Fundamentally, our current approaches are not designed to address root causes, which I believe is largely due to our limited understanding of the properties that underpin model behavior. I am interested in developing methods that leverage insights gained through understanding-driven research via two avenues:

- **Science of Language Models**: Investigating fundamental model properties under *in vitro* testbeds.
- **Holistic Evaluation**: Designing frameworks to measure higher-order capabilities.

1. Science of Language Models

I am interested in two approaches to principled investigation: (1) a theory-driven approach centered on analyzing model properties under mathematical frameworks, and (2) an experimental approach using idealized test environments.

Theory-Driven Analysis. I studied word embeddings (e.g., Word2Vec, GloVe) and their analogy parallelism, where vector representations encode linguistic analogies as parallelograms, under the guidance of **Professor Nakul Verma**. This culminated in two practical contributions: one, a significantly more train-efficient word embedding model (50× speedup), and two, a result that any well-behaved linear projection of the co-occurrence matrix yields embeddings with analogy parallelism. In my first paper [1], I analyzed the training dynamics of word embedding models and demonstrated that they inherently employ a max-margin push-and-pull objective as their underlying mechanism. I experimentally demonstrated that optimizing this objective is sufficient to yield competitive embeddings, and designed a more efficient model. However, this result did not explain the success of earlier Vector Space Models (VSMs; sparse representations that directly use co-occurrence counts as entries) in solving word analogies. In my second paper [2], I address this by introducing a probabilistic model for co-occurrence and analyzing the structure induced in high-dimensional representations. I derived bounds on the recovery rate of parallelograms between word pairs and showed that VSMs exhibit analogy parallelism under mild assumptions. This demonstrates that analogy parallelism arises from rich statistical structures of language, and even simple mappings (e.g., random projections) preserve this property.

Controlled Experiments. I developed an interest in principled testing through my project [3] under **Professor Daniel Hsu**, which investigated the emergence of in-context learning in transformer models with linear regression. Previous studies on this testbed use the random design setting, which makes disentangling data-specific and model-specific influences difficult. I addressed this by adopting the normal means problem, a statistical estimation task that reduces data-specific variability. In-context learning can also be conveniently framed as a generalizing estimator which enables easier detection, and I experimentally demonstrated that in-context learning emerges as an artifact of limited model capacity in this setting.

The aforementioned projects demonstrate that well-executed theory yields concrete findings with strong practical implications. Moving forward, I seek to combine these directions through *observation-based* research: studying model behavior on well-designed testbeds, and interpreting results via rigorous analysis.

Note, however, that such experiments are often conducted with highly controlled models and artificial constraints that abstract away the complexities of real-world LLMs. We can complement this with black-box behavioral testing of LLMs to gain immediately useful practical insights. This motivates the next section.

2. Holistic Evaluation

Robust evaluation of LLM capabilities is a growing field with many unanswered questions. The primary challenge is determining *what* to measure and *how* to measure it in a way that yields universally applicable conclusions. I have explored this through the following projects.

LLM Explainability. In a project on evaluating the precision of LLM-generated explanations under **Professor Kathleen McKeown**, we designed a new metric that quantifies the faithfulness of model-generated explanations [4]. High explanation precision is essential for reliability, as imprecise explanations can misrepresent a model's internal reasoning. To measure this, we assess whether an entity with strong reasoning skills (e.g., a human) can predict a model's counterfactual response based on its reasoning for a given question. Using this metric, we found that state-of-the-art LLMs struggle to generate precise explanations; more importantly, simply training models on human preferences does not improve explanation precision.

Jailbreaking LLMs. The above project highlighted the importance of identifying high-level, practical model behaviors. This finding motivated me to investigate high-stakes settings like "jailbreaking" LLMs to bypass safety training, in collaboration with Professor Marzyeh Ghassemi at MIT. While existing jailbreaks measure success by eliciting any response other than refusal, I developed a metric that quantifies whether a response could enable a person to enact harmful actions [5]. Using this metric, we found that truly harmful outputs can be easily generated through multi-step interactions in multiple languages, highlighting the brittleness of current preference tuning paradigms for safety.

More broadly, in tandem with developing frameworks for assessing LLM capabilities, addressing the lack of accurate metrics for practical tasks in NLP is equally important. Furthermore, task-specific evaluation can provide valuable insights: for example, measuring input coverage (how well LLMs attend to the full input) and faithfulness (how well outputs align with the input) in summarization provides a suitable testbed for investigating position bias in LLMs [6], a topic I am currently working on under **Professor Kathleen McKeown**.

3. Next Steps

To summarize, I am invested in developing fundamental theories that translate into improved future systems. I believe this is necessarily a long-term endeavor, which a PhD provides the ideal environment to pursue. Ultimately, I aspire to become a professor and combine research autonomy with the opportunity to mentor future scholars. Serving as a teaching assistant five times showed me that meaningful learning often occurs beyond the classroom.

At Princeton, I am eager to pursue both research directions under Professors **Sanjeev Arora** and **Danqi Chen**, with a slight emphasis on the first with Professor Arora and the second direction with Professor Chen. I also look forward to potential collaborations with Professors **Peter Henderson** on covert safety loopholes and **Karthik Narasimhan** on multi-agent collaboration. All in all, I believe Princeton facilitates an excellent environment to pursue my research interests.

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* *indicates co-first authorship (name reordered).*

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